

GUIDE FOR ASTM PIPING MATERIAL SPECIFICATION-- METALLIC PIPING.- ASTM Standards.

Basic Mat. Of Const.	Ch.Comp	Pipes		S W Fittings		B W Fittings		Flanges		Castings	Temp.Limit	Notes
		Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Mat.Std	Deg F(DegC)	
Carbon Steel	C	B 36.10M	A 53Gr.A/B	B 16.11	A 105	B 16.9	A 234 Gr.WPB/C	B16.5/16.47	A 105	A 216 Gr.WCA WCB, WCC	-20 to 800	1,2,3,4,5,6 10,21
			A106Gr.A/B/C								(-30 to 427)	
			A671/ A672									3,4,5,6, 10,16, 21
Low Temp.CS	C,Si	B 36.10M/ API 5L	API 5LGr. A25,P/A/B		A 350Gr.LF2 Class-1		A 420 Gr. WPL6		A 350 Gr.LF2 Class-1	A 352 Gr. LCB	- 50 to 800	3,4,5 ,17.
			A333 Gr.6								(-45 to 427)	
Low & Int.Alloy Steel for Low temp.Service	3½Ni	B 36.10M	A333 Gr.3	B 16.11	A350Gr.LF3 Class-1/2	B16.9	A 420 Gr. WPL3	B 16.5/16.47	A350 Gr.LF3	A352 Gr.LC3	- 150 to 800	3,4,5, 17.
											(-100 to 427)	
	¾ Cr.¾ Ni		A333 Gr.4		A350Gr. LF787(CI.2/3)		No Eq. Fitting Sp.		A350Gr LF787	No Eq Cstg Spec.	See Note # 11 .- 150 to 650	
											(-100 to 340)	
	2½ Ni		A333 Gr.7		No Eq Frg. Spec		No Eq Fitting Sp.		No Eq Frg.Spec	A352 Gr.LC2	- 100 to 800	
											(- 73 to 427)	
	9 Ni		A333 Gr.8		A522Type.1		A 420 Gr. WPL8		A 522 Type.1	A352 Gr.LC9	- 320 to 200	
											(-195 to 95)	
	2Ni,1Cu		A333 Gr.9		A350Gr.LF9		A 420 Gr. WPL9		A350 Gr.LF9	No Eq Cstg Spec	- 100to100 (-73to40)	
High Strength Carbon /Low Alloy Steel.	C,Mn,Co,V, Ti	B 36.10M/ API 5L	API 5L X42	B 16.11	A694 Gr. F42	B 16.9	A860 Gr. WPHY 42	B 16.5/16.47	A694 Gr. F42	A216Gr.WCC	- 20 to 400	5,19
											(-30 to 204)	
	(All API 5L PSL 2 Pipe)		API 5L X46		A694 Gr. F46		A860 Gr. WPHY 46		A694 Gr. F46	No Eq Cstg Spec	-20 to 400 (-30 to 204)	
			API 5L X52		A694 Gr. F52		A860 Gr. WPHY 52		A694 Gr. F52	A217Gr.CA15	-20 to 400	
											(-30 to 204)	
			API 5L X56		A694 Gr. F56		A860 Gr. WPHY 60		A694 Gr. F60	A217Gr.C5	-20 to 400	
											(-30 to 204)	
			API 5L X60		A694 Gr. F60		A860 Gr. WPHY 60		A694 Gr. F60	A217Gr.C5	-20 to 400	
											(-30 to 204)	
			API 5L X65		A694 Gr. F65		A860 Gr. WPHY 65		A694 Gr. F65	A217Gr.CA15	-20 to 400	
											(-30 to 204)	
			API 5L X70		A694 Gr. F70		A860 Gr. WPHY 70		A694 Gr. F70	No Eq. Cstg Spec		

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High Strength Carbon /Low Alloy Steel.	(All API 5L PSL 2 Pipe)	B 36.10M/ API 5L	API 5L X80		No Equivalent Forging Spec	MSS-SP-75.	Material Same as Pipe.		No Equivalent Forging Spec.	No Equivalent Casting Spec	-20 to 400	
			API 5L X90								(-30 to 204)	
			API 5L X100									
			API 5L X120									
Low & Int. Alloy Steel for High temp Service	C, ½ Mo	B 36.10M	A335 Gr.P1	B 16.11	A182 Gr.F1	B 16.9	A234Gr.	B16.5/16.47	A182 Gr.F1	A217Gr.	-20 to 900	3,4,5,7
			A691CM65				WP1			.WC1	(-30 to 482)	
	½ Cr, ½ Mo		A335 Gr.P2				No Eq		A182 Gr.F2	A217 Gr.	-20 to 900	
	(Ni,Cu,V+Al)		A335 Gr.P36				Fitting Sp.		A182 Gr.F36	WC4	(-30 to 482)	-
			A691 ½ CR									
	1Cr, ½ Mo		A335 Gr.P12				A234 Gr.		A182 Gr.F12	No Eq.	-20 to 950	3,4,5,7,12
			A691 1 CR				WP12			Cstg.Spec.	(-30 to 510)	
	1¼ Cr,		A335 Gr.P11				A234 Gr.		A182 Gr.F11	A217Gr.	-20 to 900	3,4,5,7,12
	½ Mo		A691 1¼ CR				WP11 Cl.2		Cl.2	WC6	(-30 to 482)	
	1½ Si,		A335 Gr.P15				No Eq		No Eq	A217Gr.	-20 to 950	
	½ Mo						Forg. Sp.		Forg. Sp.	CA15	(-30 to 510)	
	2¼ Cr,1Mo		A335 Gr.P22				A234 Gr.		A182 Gr.F22	A217Gr.	-20to950	3,4,5,7,12
	(V,Cb,W,B,N)		A335 Gr.P23				WP22 Cl.1		Cl.1	WC9	(-30to510)	
			A691 2¼ CR									
	3Cr,1Mo		A335 Gr.P21				No Eq		A182 Gr.F21	No Eq	-20to900	3,4,7,12
			A691 3 CR				Fitting Sp.			Cstg.Spec	(-30to482)	
	5Cr, ½ Mo		A335 Gr.P5				A234 Gr.		A182 Gr.F5	A217Gr.	-20 to 900	3,4,5,7,12
			See Note#12				WP5			C5	(-30to482))	
			A691 5 CR									
	9Cr,1Mo		A335 Gr.P9				A234 Gr.		A182 Gr.F9	A217Gr.	-20to950	3,4,5,7,12
			A691 9 CR				WP9 Cl.1			C12	(-30to510)	
	9Cr,1Mo,V		A335 Gr.P91				A234 Gr.		A182 Gr.F91	A217Gr.	-20to1150	3,4,5,7,12
			A691 91				WP91			C12A	(-30to610)	
	9Cr,2W,½ Mo		A335 Gr.P92						A182 Gr.F92	No Cstg.Sec		
	11Cr,½ Mo		A335 Gr.P122				No. Ftg Spec.		A182 Gr.F122	No Cstg.Sec		
							A234 Gr.					
	9Cr.1Mo,W		A335Gr.P911				WP911		A182 Gr.F911	No Cstg.Sec		Sheet 2/ 6

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		Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Mat.Std	Deg F(DegC)	
Austenitic Stainless Steel	18Cr,8Ni	B 36.19M	A312 TP 304		A182 Gr.F304	B 16.9/	A403 Gr.	B 16.5/16.47	A182 Gr.F304	A351Gr.CF8	See Note #13	
			A358 Gr.304			MSS-SP43	WP304 CR304					3,4,5,8,9
	18Cr,8Ni, Low C		A312 TP 304L A358 Gr.304L		A182 Gr. F304L		A403 Gr. WP304L CR304L		A182 Gr. F304L	A351Gr.CF3		3,4,5,8,9
	18Cr,8Ni, 3Mo		A312 TP 316 A358 Gr.316		A182 Gr.F316		A403 Gr. WP316 CR316		A182 Gr.F316	A351Gr. CF8M		3,4,5,8,9
	16Cr11Ni, 3Mo, LowC		A312 TP 316L A358 Gr.316L		A182 Gr. F316L		A403 Gr. WP316L CR316L		A182 Gr. F316L	A351Gr. CF3M		3,4,5,8,9
	16Cr,11Ni, Ti Stab.		A312 TP 321 A358 Gr.321		A182 Gr.F321		A403 Gr. WP321		A182 Gr.F321	No Eq. CstgSpec.		3,4,5,8,9
	18Cr,8Ni, 3Mo,Ti Stab.		A312 TP316Ti		A182 Gr. F316Ti		No Eq. fitting Spec.		A182 Gr. F316Ti	No Eq. Cstg Spec.		3,4,5,8,9
	18Cr,11Ni. Cb Stab		A312 TP 347 A358 Gr.347		A182 Gr.F347		A403 Gr. WP347 CR347		A182 Gr.F347	A351Gr. CF8C		3,4,5,8,9
	25Cr,20Ni		A312 TP 310 A358 Gr.310		A182 Gr.F310		A403 Gr. WP310 CR310		A182 Gr.F310	A351Gr. CK20		3,4,5,8,9
	19Cr11Ni, 3½ Mo		A312 TP317 A358 Gr.317		A182 Gr.F317		A403 Gr. WP317 CR317		A182 Gr.F317	A351Gr. CG8M		3,4,5,8,9
	21Cr,25Ni, 6½ MoLoC		A312 N08367 A358 N08367		A182 Gr.F62		A403 Gr. WP6XN CRWPXN		A182 Gr.F62	A351Gr. CN3MN		3,4,5,8
	21Cr,26Ni, 4½ MoLoC		A312 N08904 A358 N08904		A182 Gr. F904L		A403 Gr. WP904L CR904L		A182 Gr. F904L	No Eq. Cstg Spec.		3,4,5,8

Special Notes:-

- 1) Grades **UNS N08367,N08904, N08925 & N08926** were previously covered under non-ferrous category ,now recognized as Austenitic Stainless Steel.
- 2) Normal Grade has **0.08% C**, Grade **H** has **0.04 to 0.1 % C** and Grade **L** has **0.035% C** and is of lower strength.

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		Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Mat.Std	Deg F(DegC)		
Ferritic/Austenitic	22Cr,5½ Ni	B 36.19M	A790	B 16.11	A182 Gr.F51	B 16.9/	A815	B 16.5/16.47	A182 Gr.F51	A 890 Gr.4A	-60 to 600	3,4,5,8,20	
(Duplex)	3Mo		S 31803				S 31803			(UNS J92205)	(-50 to 315)		
Stainless Steel							CRS31803.						
	23Cr,4Ni,N		A 790				A182 Gr.F68		No Eq.	A182 Gr.F68	A 890 Gr.1B	-60 to 600	3,4,5,8,20
			S 32304						Fitting.Spec.		(UNSJ93372)	(-50 to 315)	
			(Type 2304)										
	26Cr, 4Ni,Mo		A 790				No Eq.		No Eq.	No Eq.	A 890 Gr.3A	-20 to 600	3,4,5,8,20
			S 32900				Forging.Spec.		Fitting.Spec.	Forging.Spec.	(UNSJ93371)	(-30 to 315)	
			(Type329)										
	26Cr,3½ Ni		A 790				A182 Gr.F52		A 815	A182 Gr.F52	No Eq.	-60 to 600	3,4,5,8,20
	1Mo		S 32950						S 32950		Cstg.Spec.	(-50 to 315)	
									CRS32950.				
	25Cr,7Ni,		A 790				A182 Gr.F53		A 815	A182 Gr.F53	A 890 Gr.5A	-20 to 600	3,4,5,8,20
	4Mo,N		S 32750						S 32750		(UNSJ93404)	(-30 to 315)	
			(TYPE2507)						CRS32750.				
	25Cr,7Ni,		A 790				A182 Gr.F54		A 815	A 182 Gr.F54	A 890 Gr.6A	-60 to 600	3,4,5,8,20
	3Mo,N,W,Cu		S 39274						S 39274		(UNSJ93380)	(-50 to 315)	
									CRS 32974				
	25Cr,7Ni		A 790				A182 Gr.F55		A 815	A182 Gr.F55	A 995 Gr.6A	-60 to 600	3,4,5,8,20
	3Mo,W,Cu		S32760						S32760		CD3MWCuN	(-50 to 315)	
									CRS32760		(UNS J93380)		
	22Cr,5½ Ni		A 790				A182 Gr.F60		A 815	A182 Gr.F60	A 890 Gr.4A	-60 to 600	3,4,5,8,20
	3Mo,N		S 32205						S 32205		(UNS J92205)	(-50 to 315)	
									CRS 32205				
	26Cr,6Ni		A 790				A182 Gr.F61		A 815	A182 Gr.F61	A 890 Gr.1C	-60 to 600	3,4,5,8,20
	3½ Mo,N,Cu		S 32550						S 32550		(UNS J93373)	(-50 to 315)	
					CRS 32550								
29Cr,6½ Ni	A 790		A182 Gr.F65		No Eq.		A182 Gr.F65		No Eq.	-60 to 600	3,4,5,8,20		
2Mo,N,Cu	S 32906	Fitting Sp.		Cstg.Spec.	(-50 to 315)								

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		Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Dimn.Std	Mat.Std	Mat.Std	Deg F(DegC)	
NON-FERROUS MATERIALS												
Titanium	Unalloyed Ti	B 36.19M/ B 36.10M	B861/B862	B 16.11	B381	B 16.9/ MSS-SP43	B363 Gr.WPT1	B16.5 Appendix-A	B381	B367	-75 to 600 (-60 to 315)	4,5,8,13, 14, 24
Inconel-825 (UNS N08825)	42Ni,22Cr, 22Fe,3Mo	B 36.19M	B 423 / B 705 UNS N08825	B 16.11	B 564 UNS N08825	B 16.9/ MSS-SP43	B 366 WPNiCMC CRNiCMC	B 16.5/16.47	B564 UNS N08825	A494 Gr. Cu5MCuC	-320 to 1000 (-195 to 538)	3,4,5,8
Inconel-800 (UNS N08800)	33Ni,21Cr, 40Fe	B407 Table X1.1	B 407 UNS N08800	B 16.11	B564 UNS N08800	B 16.9/ MSS-SP43	B 366 WPNiC CRNiC	B 16.5	B564 UNS N08800	A351Gr. CT15C	-325 to 1500 (-198 to 816)	5,8,15
Inconel-625 (UNS N06625)	60Ni,22Cr, 9Mo,4Nb	B 36.19M	B 444 / B 705 UNSN06625	B 16.11	B 564 UNS N06625	B 16.9/ MSS-SP43	B 366 WPNCMC CRNCMC	B 16.5	B 564 UNSN06625	A 494 Gr. CW6MC	-325 to 1200 (-198 to 649)	4,8
Inconel-600 (UNS N06600)	72Ni,15Cr, 8Fe	B 36.19M	B167/ B517	B 16.11	B 564 UNS N06600	B 16.9/ MSS-SP43	B 366 WPNCI CRNCI	B 16.5	B 564 UNS N06600	A 494 Gr. CY40	-325 to 1200 (-198 to 649)	4,8
Hastalloy-C (UNS N10276)	15Cr,16Mo 5Fe,57Ni	B 36.19M	B 622 / B 619 UNS N10276	B 16.11	B 462 UNS N10276	B 16.9 MSS-SP43	B 366 WPHC276 CRHC276	B 16.5	B 462 UNS N10276	A 494 Gr. CW12MW	-325 to 1250 (-198 to 677)	4,8
										UNS N30002		
Hastalloy-B (UNS N10665)	15Cr,16Mo 5Fe,57Ni	B 36.19M	B 622 / B 619 UNS N10665	B 16.11	B 462 UNS N10665	B 16.9 MSS-SP43	B366 WPHB-2 CRHB-2	B 16.5	B 462 UNS N10665	A494 Gr. N3M UNS J30003		
Alloy-20 (UNS N08020)	35Ni,20Cr Cb,Fe	B 36.19M	B 729 / B464 UNS N08020	B16.11	B 462 UNS N08020	B 16.9 Notes 5&8	B 366 CR20CB	B 16.5	B 462 UNS N08020	A 351 Gr. CN7M	-325 to 800 (-198 to 426)	4,8
Nickel (UNS N02200) (UNS N02201)	Ni	B 36.19M/	B161 / B725 (UNS N2200)	B16.11	B 564	B 16.9	B366 WPN B366 CRN	B 16.5	B 564	A 494 Gr.	-325 to 1200	4, 8,
	LC-Ni	B 36.10M	B161 / B725 (UNS N2201)		B 160	Notes 5 & 8	B366 WPNL B366 CRNL		B160	CZ100	(-198 to 649)	
Monel-400 (UNS N04400)	63Ni,30Cu	B165 Table X2.1	B165 / B725	B16.11	B564	B 16.9 See Note# 5	B 366 WPNC CRNC	B 16.5	B 564 UNS04400	No Eq.Cstng.	-325 to 850 (-198 to 482)	4, 8,
Aluminum	Al	ANSI H35.2	B241 / B345	B16.11	B 361	B 16.9 See Note# 5	B 361	B16.5 Appendix-A	B 247	B 618	-452 to 300 (-270 to 315)	13, 24
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Notes:- (All Material standards are as per **ASTM** and Dimensional Standards are as per **ASME**.)

- 1) A671 & A672 are plate fabricated pipes $\geq$ NPS16. Grade and Class as per plate mat. For A671 there are 28 Classes and for A672 there are 20 Classes.
- 2) Pipes to ASTM A 106 is widely used for services with moderate & high temperatures up to NPS 14 normally within the prescribed limit of temp.
- 3) ASME B 16.5 covers flanges from NPS $\frac{1}{2}$ (DN15) to NPS 24 (DN600). For flanges from NPS 26 to NPS 60 refer ASME B 16.47. Select Series **A** or **B**
- 4) ASME B 16.11 covers Socket Weld / Screwed fittings up to NPS 4 (DN 100). Generally used only upto **DN40**.
- 5) ASME B 16.9 covers BW fittings up to NPS 48(DN1200). To specify fittings beyond this size, add suffix "**S58**" to the material spec. (See A960)
If dimensional requirements to B16,9 are not exactly followed, then add suffix "**S9**" to ferrous and "**SPLD**" to non-ferrous fittings.(See Clause 4.4.2)
MSS-SP-75 covers BW fittings from NPS 16 to NPS 60 in Carbon Steel and low & intermediate Alloy Steel. Up to NPS 14 refer ASME B16.9.
MSS-SP-43 covers fittings of SS Grades 304,304L,316,316L & 347 in NPS 1/2 to NPS24 in Sch. 5S & Sch. 10S thk. and are of "**CR**" category only.
- 6) Selection of grades to match mechanical strength with that of Fittings/Castings.
- 7) Pipes to A 369 can also be selected but it is a forged and bored pipe..
- 8) In case of Stainless Steel / Non-ferrous Fittings, Seamless construction to be specified as **WPS** and with Welded construction to be specified as **WPW**.
If welds are UT tested, then the designation to be **WPWU** and if Radiographically tested it should be **WPWX**. No Class & Tests for **CR** Grades.
In case of Carbon and Alloy Steels **WP** stands for Seamless ,add suffix **W** for welded quality and **WU** for Ultrasonic testing in lieu of Radiography..
- 9) A358 is plate fabricated pipe. For class refer standard. Pipes to A 409 can also be selected.A409 covers EFW pipes NPS 14 to 30 in Sch. 5S &10S thk.
- 10) A53 Gr. **A** ,API 5L Grade **A25** and **X42** and above are not recommended by Code to be used above 400° F
- 11) A 350 Gr. LF787 Cl.2 is up to -75°F(-59°C) and Cl.3 is up to -100°F(-73°C) whereas A 333 Gr. 4 is suitable for -150°F to 650°F(-100°C to 345°C)
- 12) A 335 Gr. P5 has three grades P5, P5b and P5c, A182 Gr.F11& A234 WP11has 3, F12./WP12 has 2 & F22/WP22 has 2 Classes each.
- 13) Aluminum can be used for Cryogenic Service up to — 452°F(—270°C)and Austenitic Stainless Steel up to — 425°F(—255°C)
- 14).B 337 covers pipe sizes up to NPS 12 (DN 300) and B 861/ 862 cover sizes up to NPS 36 (DN900)
- 15) B407 & B 466 cover pipes up to NPS 6 (DN150) only.
- 16) For API 5L, two Product Specification Levels - PSL-1 & PSL-2 & Grades are **A25, A, B, X42, X46, X52, X56, X60, X65, X70, X80, X90, X100 & X120**.
PSL-1 is available in Grades fro A25 to X70 and PSL-2 from Gr.B to X120. Gr.A25 is limited to NPS 5.Through Flow Line (TFL) pipes are furnished in double random lengths.
- 17) To be impact tested to the specified Minimum Temperature. Gr.10 (—75°F) & 11(—320°F) are listed in A333, but not adapted by B31.3 yet.
- 18) Materials grades listed in ASME **B 31.3** are mainly considered along with other popular grades in this Chart. Though other grades are also available as per the Material Standards.
- 19) Grade from Casting Specification is selected to match the mechanical strength of the pipe with the cast component
- 20) ASTM A 890 is the Duplex SS casting for General application and A 995 is that for pressure containing parts.
- 21) Heat Treatment not mandatory for flanges in Class 150 & 300.When heat treatment to be done by normalizing, material is designated as 105N.
- 22) For the selection of materials for Hydrogen service at high temperature & high pressure, use NELSON Curve in API RP941.
- 23) For selection of Piping materials for Sweet / Sour service, requirements outlined in NACE standard MR 0175/ 0103 shall be followed.
To select MOC for Caustic Service refer '**Caustic Service Chart**' published by NACE in SP 0403 & for Concentrated H2SO4, refer NACE RP0391
- 24) Titanium and Aluminum are not listed in ASME B16.5. Use Appendix A to establish flange rating.